CHAPTER 1

Introduction

The rapid growth of urbanization and increasing population density in local government units (LGUs) across the Philippines has intensified the frequency and complexity of human-induced incidents in communities. Road accidents, fires, criminal activities, infrastructure damage, and public disturbances are among the most common emergencies that affect the daily lives of citizens and challenge the response capabilities of local authorities. The timely detection, reporting, and resolution of these incidents are critical in reducing harm, minimizing damage, and maintaining public safety (Shkedova, 2022).

Traditional incident management practices in many Philippine barangays, municipalities, and cities still rely on manual processes, verbal communications, and fragmented reporting methods. These approaches are often prone to delays, inaccuracies, and limited visibility into the real-time status of ongoing incidents. Response units are frequently dispatched without precise location data, and administrators lack the tools to monitor incident trends, identify high-risk areas, or evaluate response efficiency over time. Advances in information technology, particularly in the fields of Geographic Information Systems (GIS), web-based applications, and data analytics, have created new opportunities to modernize incident management at the local level (Daud, 2025). Several studies and working systems have demonstrated that integrating map-based visualization, real-time dashboards, automated notifications, and spatial analysis into a single platform

significantly improves the speed, accuracy, and coordination of emergency response operations (Shkedova, 2023).

In response to these challenges and opportunities, the researchers propose the development of a Geospatial Approach to Optimize Incident Response System with Data Analytics Integration. This web-based system aims to provide barangay officials, response units, and administrators with a unified platform for reporting, monitoring, and analyzing human-induced incidents within a defined local jurisdiction. By combining GPS-based incident reporting, interactive geospatial mapping, geofencing, nearest responder detection, automated alerts, and a real-time monitoring dashboard with hotspot and trend analysis, the proposed system seeks to fill the critical gaps in current local incident management practices.

Background of the Study

Incident management at the barangay and municipal level in the Philippines is a persistent operational challenge. Local government units are mandated to protect community safety and coordinate emergency responses, however, many LGUs lack the technological infrastructure to manage incidents efficiently. The Disaster Risk Reduction and Management (DRRM) framework encourages local preparedness, yet the tools available for day-to-day human-induced incident monitoring remain largely underdeveloped at the community level. Existing studies highlight the effectiveness of geospatial tools in improving emergency response. A study on emergency

incidents and disaster response management in Oton, Iloilo identified significant gaps in service coverage and response efficiency when incidents were not analyzed through spatial and temporal lenses (Aporto, 2026). Similarly, a spatiotemporal study of fire incidents in Pampanga demonstrated that mapping incident patterns enables authorities to plan targeted prevention and response programs (de Leon & Miranda, 2022). These findings underscore the practical value of geospatial analysis in local emergency operations.

Locally developed systems such as iAlerto for Pasig City DRRMO, the "Help Me" emergency response application in Dingle, Iloilo, and the Video-Based Fire Detection and Reporting System demonstrate that web and mobile platforms with GPS integration and automated alerts are feasible and acceptable solutions for local emergency reporting. However, these systems tend to address isolated components of incident management rather than providing a fully integrated workflow that covers reporting, response coordination, and analytical monitoring in a single platform. The proposed system is designed to bridge this gap. By anchoring its design in established geospatial methodologies and drawing from both local and international research and system literature, it aims to deliver a comprehensive, role-based incident management platform tailored to the operational realities of Philippine local government units.

Objectives of the Study

The general objective of this study is to design and develop a Geospatial Approach to Optimize Incident Response System with Data Analytics Integration that enables efficient reporting, coordinated response, and comprehensive monitoring of human-induced incidents within a local jurisdiction.

Specifically, the study aims to:

1. Develop an Incident Reporting Module that allows authorized users to submit incident reports with GPS coordinates, photo evidence, incident type, and descriptive details through a web-based digital form.
2. Implement a geospatial mapping interface that plots reported incidents on an interactive map using color-coded markers categorized by incident type, and applies geofencing to assign incidents to the correct barangay or zone.

3. Build an Incident Response Module that automatically calculates the nearest available response unit to a reported incident and sends automated alerts through the system dashboard, SMS, email, or mobile notification.
4. Create a real-time Monitoring and Analysis Module that provides administrators and officials with a dashboard displaying incident counts, statuses, and geographic distribution, along with heatmap and hotspot visualization tools.
5. Enable data export and report generation functionality that allows officials to produce incident summaries, area-based reports, and response time analysis in PDF, Excel, and CSV formats.
6. Evaluate the system's functionality, usability, and performance using accepted software quality standards and feedback from intended users.

Scope and Delimitation of the Study

The proposed system covers the design, development, and evaluation of a Geospatial Approach to Optimize Incident Response System with Data Analytics Integration focused on human-induced or unnatural incidents occurring within a defined local jurisdiction, such as a barangay, municipality, or city. The system encompasses three core or major modules including the Incident Reporting Module, the Incident Response Module, and the Monitoring and Analysis Module.

- Road accidents
- Fires caused by human activity
- Crime-related events
- Infrastructure damage
- Public disturbances
- Other human-induced emergencies within the system's classification scheme

The system supports role-based access for three primary user groups including administrators, authorized reporters (such as barangay officials and field personnel), and response units. Citizens may submit community reports depending on configuration, with verification workflows in place before responders are dispatched.

The system is explicitly delimited and does not cover natural disasters such as earthquakes, typhoons, floods caused by natural phenomena, landslides, volcanic eruptions, or tsunamis, as these are handled by specialized disaster monitoring systems and national agencies such as NDRRMC and PAGASA. The system is not intended to replace national emergency management infrastructure but to complement and strengthen local government response capacity. The geographic scope is limited to the jurisdiction area configured during system deployment. The system does not support cross-jurisdictional coordination or inter-LGU incident sharing in its current iteration.

Additionally, the system requires internet connectivity for real-time features and does not include offline-first functionality in this version.

Significance of the Study

The development of the proposedGeospatial Approach to Optimize Incident Response System with Data Analytics Integration is significant to multiple stakeholders within the local government and community context.

For local government units (LGUs) and barangay officials, the system provides a structured and centralized platform to oversee incident reporting, monitor response status in real time, and access data-driven insights that support evidence-based planning and resource allocation. Officials gain the ability to identify incident hotspots, track response efficiency, and generate reports that support budgeting, policy decisions, and community safety programs.

For response units such as fire stations, police units, barangay emergency teams, and disaster response personnel, the system reduces response delays by providing precise incident location data and automated notifications. The nearest responder detection feature ensures that the most appropriate unit is alerted immediately, improving dispatch efficiency and reducing the time between incident reporting and on-site response.

For community members and residents, the system enables faster emergency assistance by streamlining the reporting process and ensuring that submitted incidents are verified, logged, and acted upon through a transparent status tracking workflow. This contributes to increased public trust in local emergency services.

For researchers and future developers, the system contributes to the body of knowledge on geospatial incident management systems in the Philippine local government context. It demonstrates a practical integration of GIS, automated notification, and data analytics within a single platform, and identifies areas for future enhancement such as offline reporting, predictive analytics, and multi-jurisdiction coordination.

For the broader field of information systems and public administration, this study supports the argument that technology-driven solutions, grounded in geospatial and data analytics principles, can meaningfully improve public safety operations even at the barangay and municipal level.

Definition of Terms

The following terms are defined operationally as used throughout this study:

Barangay — The smallest administrative unit of local government in the Philippines, responsible for maintaining order and coordinating community-level emergency response.

Geofencing — A location-based feature used in the system to define virtual geographic boundaries corresponding to barangay or zone jurisdictions, enabling automatic assignment of reported incidents to the correct administrative area.

Geographic Information System (GIS) — A framework for gathering, managing, and analyzing spatial and geographic data used in this system for incident mapping, hotspot visualization, and nearest responder detection.

GPS Coordinates — Geographic data in the form of latitude and longitude values automatically captured by the system from the reporting device to identify the precise location of an incident.

Heatmap / Hotspot Visualization — A geospatial analytical feature that displays the density and concentration of incidents across geographic areas using color gradients, helping administrators identify high-risk locations.

Human-Induced Incident — Any emergency or unnatural event caused by human activity or negligence, including but not limited to road accidents, fires from human causes, criminal events, public disturbances, and infrastructure damage. This is the primary category of incidents covered by the proposed system.

Incident Status — A classification that indicates the current state of a reported incident within the system workflow. The defined status stages are: Reported, Verified, Responding, Resolved, and Closed.

Local Government Unit (LGU) — A government body at the city, municipal, or barangay level responsible for local governance, public safety, and emergency management within a defined jurisdiction.

Monitoring Dashboard — A real-time visual interface within the system that displays incident counts, statuses, geographic distribution, and key performance indicators to administrators and authorized users.

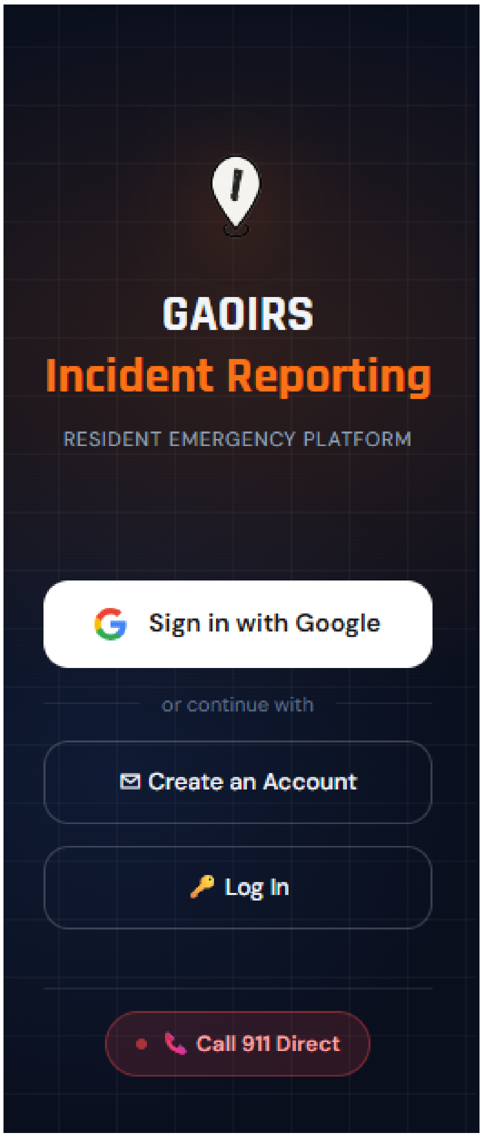
Nearest Responder Detection — A geospatial computation feature that calculates the proximity of available response units to a reported incident location and identifies the closest unit for automated alert and dispatch.

Response Unit — An authorized emergency or safety organization assigned to acknowledge and respond to reported incidents. Examples include fire stations, police stations, barangay emergency units, and disaster response teams.

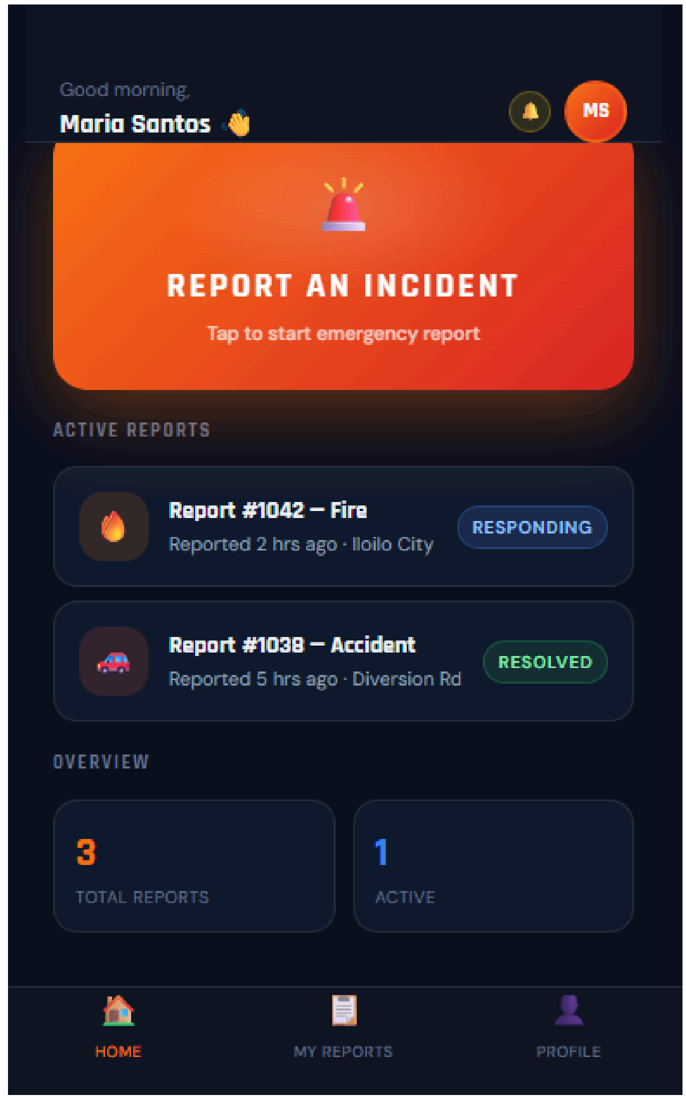
Web-GIS — A web-based implementation of Geographic Information System technology that enables geospatial data visualization, analysis, and interaction through a standard internet browser without the need for specialized desktop software.

Reporting Dashboard (Mobile App) for Residents

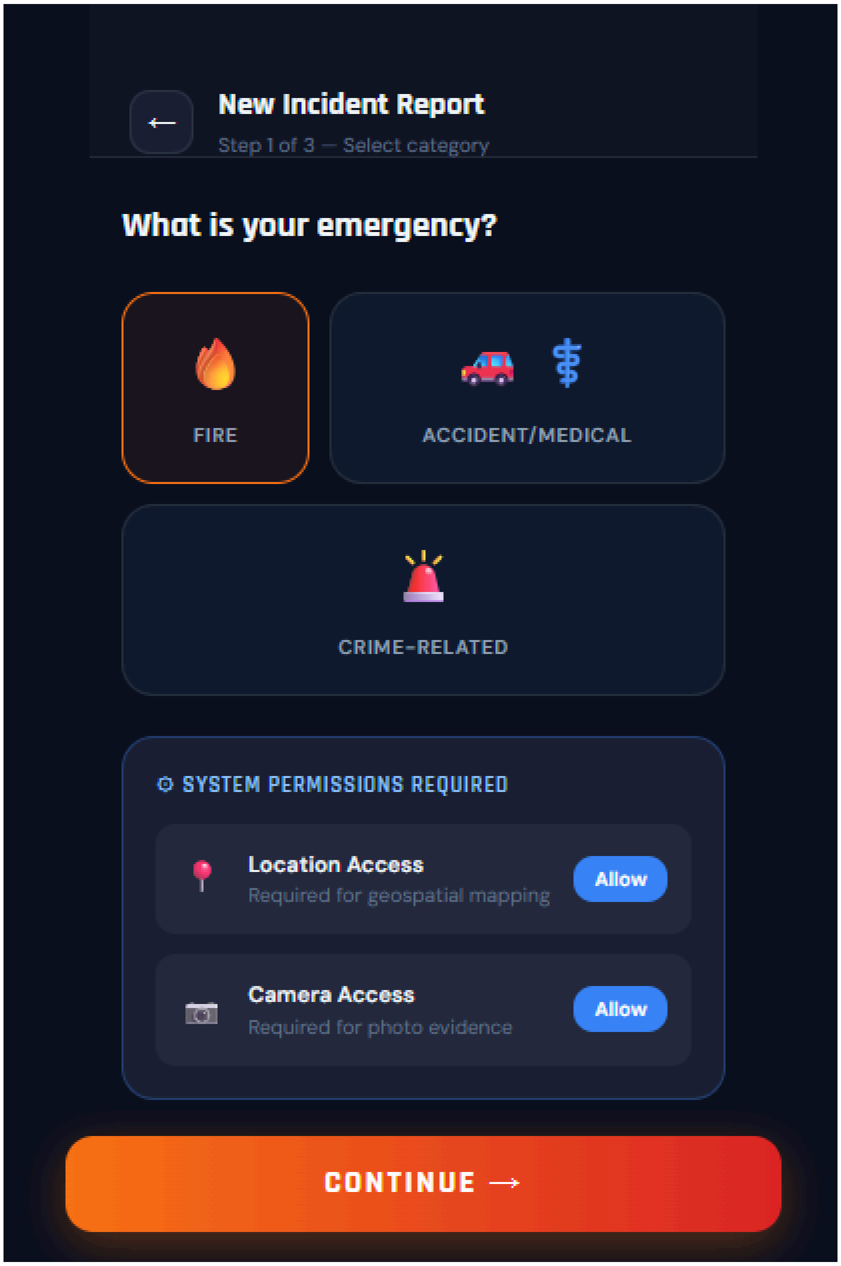
Screen 1:



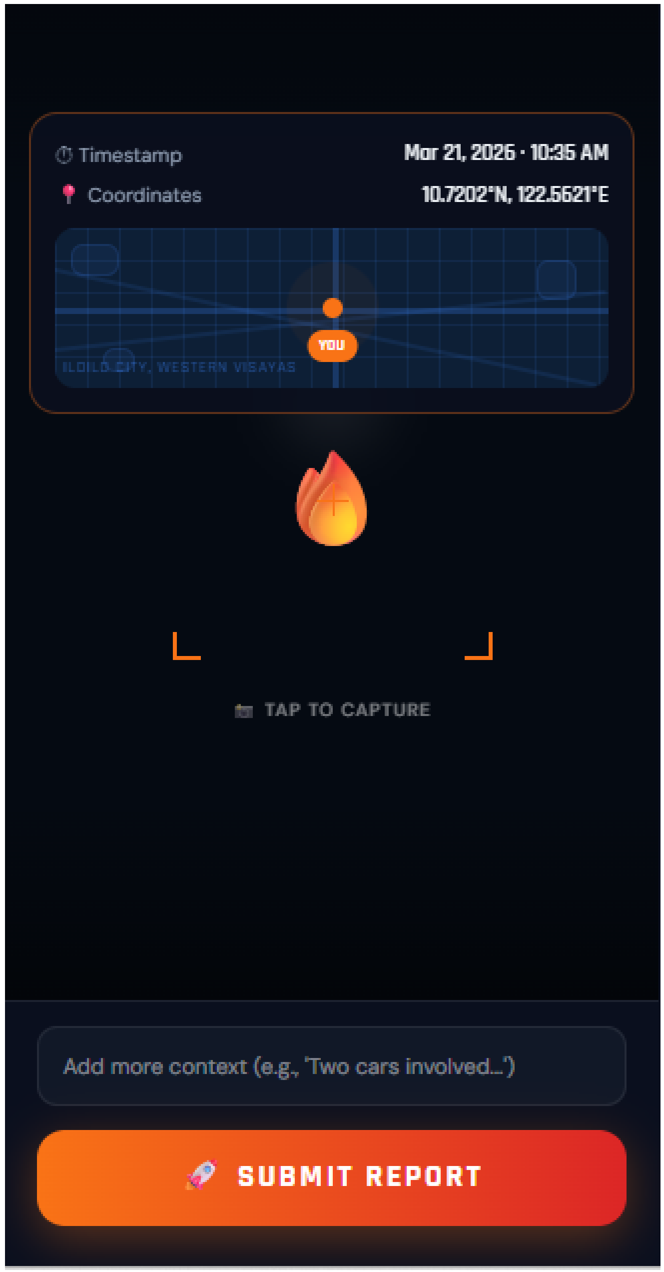
Screen 2:



Screen 3:

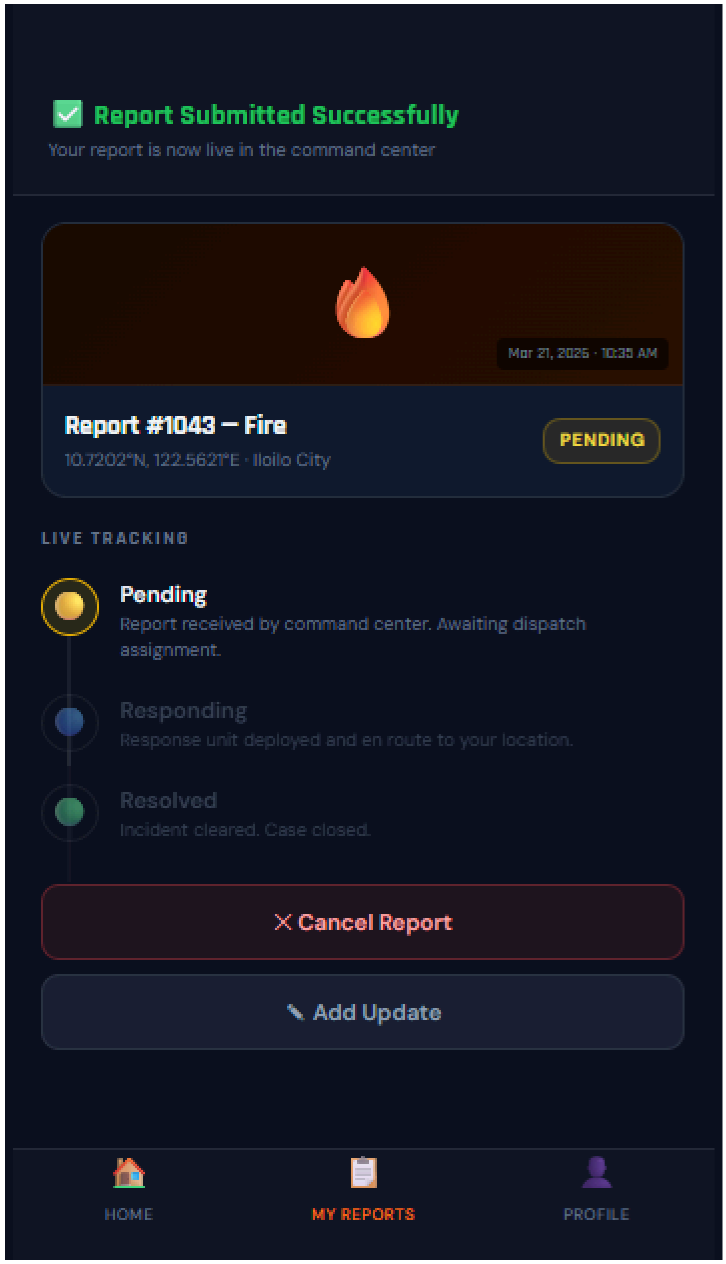

New Incident Report
Step 1 of 3 — Select category
What is your emergency?
FIRE
ACCIDENT/MEDICAL
CRIME-RELATED
SYSTEM PERMISSIONS REQUIRED
Location Access
Required for geospatial mapping
Allow
Camera Access
Required for photo evidence
Allow
CONTINUE →

Screen 4:



⏱ Timestamp
Mar 21, 2026 · 10:35 AM
Coordinates
10.7202°N, 122.5621°E
YOU
ILOILO CITY, WESTERN VISAYAS
TAP TO CAPTURE
Add more context (e.g., 'Two cars involved...')
SUBMIT REPORT

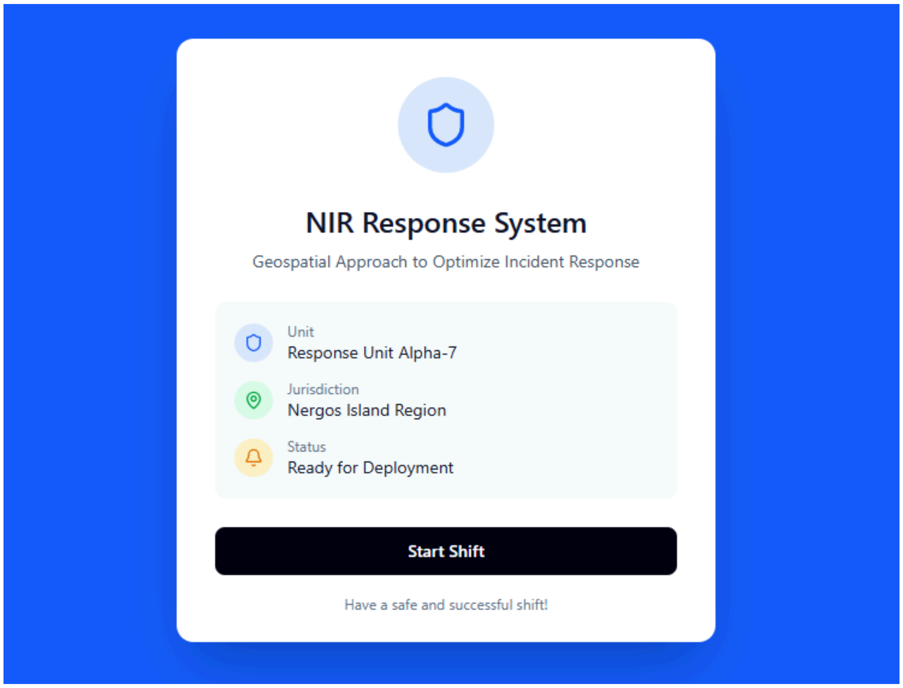
Screen 5:



Figma Prototype Link: https://figure-amity-65762716.figma.site/dashboard

Response Dashboard

Screen 1:



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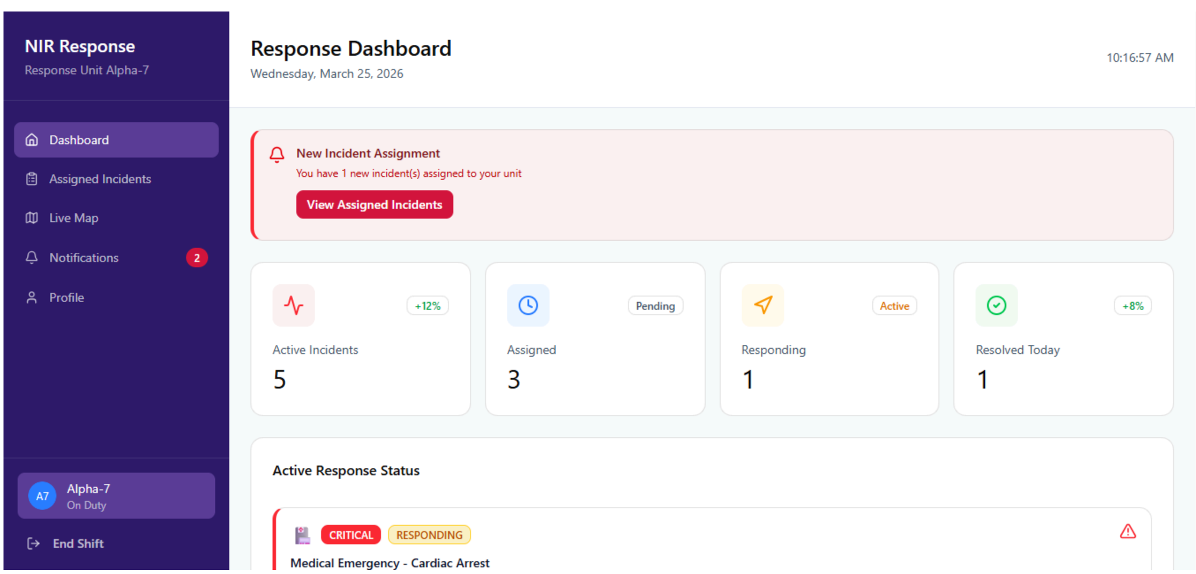
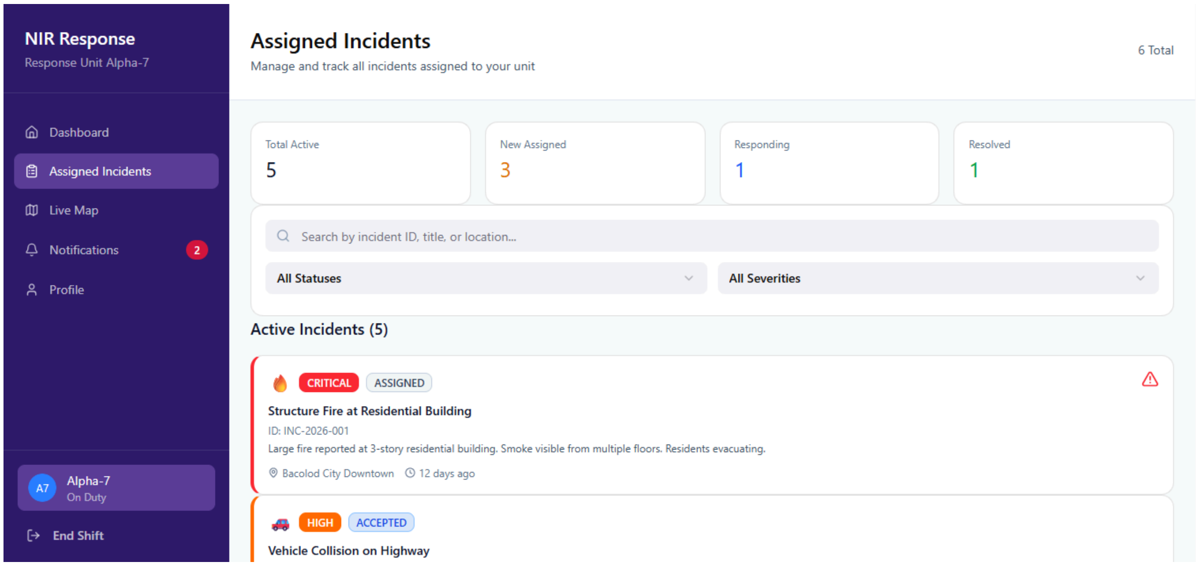
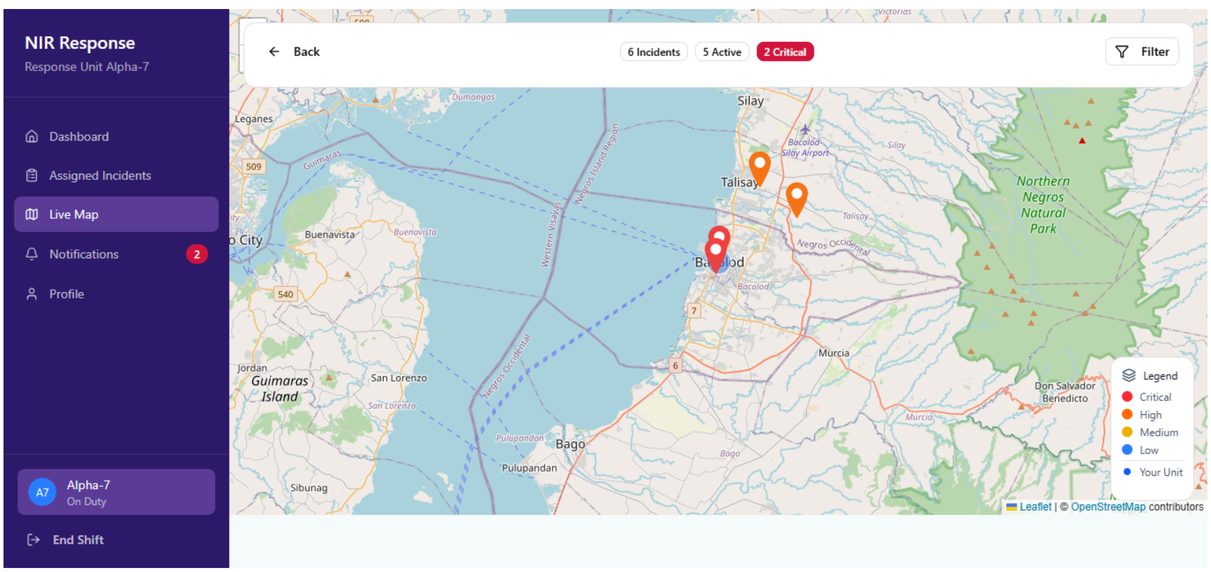


Screen 3:

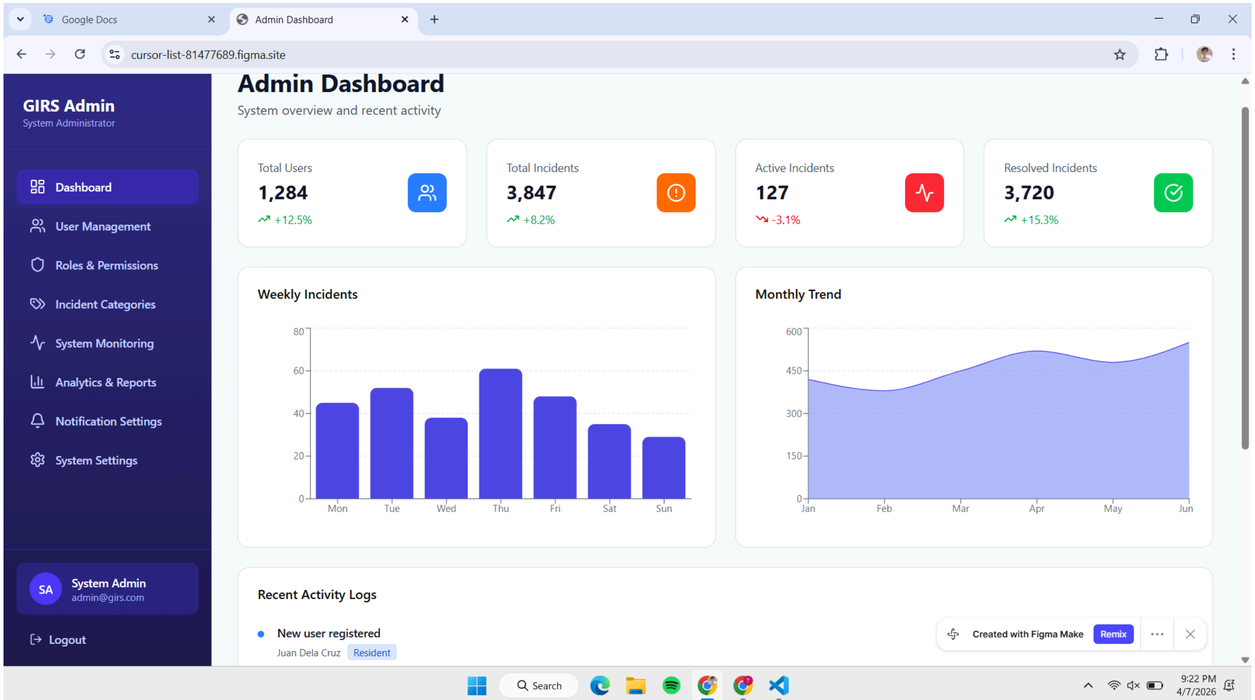


Screen 4:

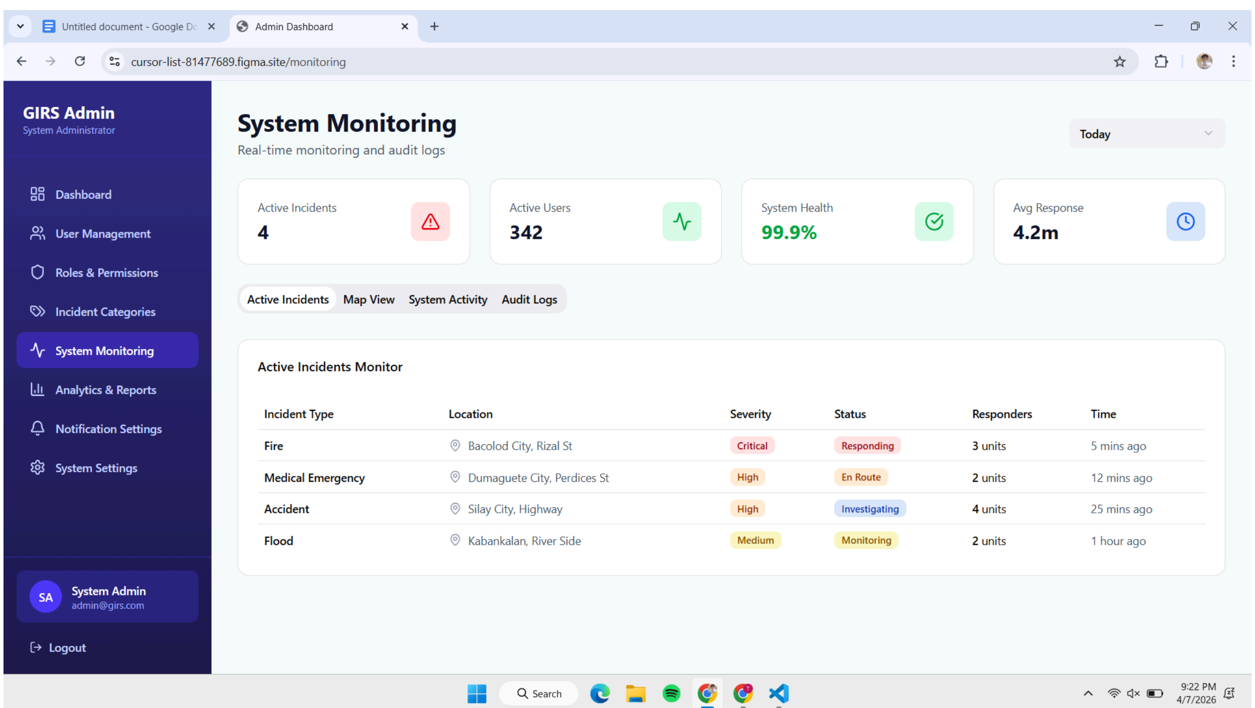


Prototype Response Link: https://mold-shaky-99788591.figma.site/map

Admin



Admin Dashboard
System overview and recent activity
GIRS Admin
System Administrator
Dashboard
User Management
Roles & Permissions
Incident Categories
System Monitoring
Analytics & Reports
Notification Settings
System Settings
Total Users
1,284
+12.5%
Total Incidents
3,847
+8.2%
Active Incidents
127
-3.1%
Resolved Incidents
3,720
+15.3%
Weekly Incidents
Monthly Trend
Recent Activity Logs
New user registered
Juan Dela Cruz
Resident
System Admin
admin@girs.com
Logout
Created with Figma Make
Remix



System Monitoring
Real-time monitoring and audit logs
Today
GIRS Admin
System Administrator
Dashboard
User Management
Roles & Permissions
Incident Categories
System Monitoring
Analytics & Reports
Notification Settings
System Settings
Active Incidents
4
Active Users
342
System Health
99.9%
Avg Response
4.2m
Active Incidents
Map View
System Activity
Audit Logs
Active Incidents Monitor
Incident Type
Location
Severity
Status
Responders
Time
Fire
Bacolod City, Rizal St
Critical
Responding
3 units
5 mins ago
Medical Emergency
Dumaguete City, Perdices St
High
En Route
2 units
12 mins ago
Accident
Silay City, Highway
High
Investigating
4 units
25 mins ago
Flood
Kabankalan, River Side
Medium
Monitoring
2 units
1 hour ago
System Admin
admin@girs.com
Logout


Untitled document - Google Docs
Admin Dashboard
cursor-list-81477689.figma.site/analytics
GIRS Admin
System Administrator
Dashboard
User Management
Roles & Permissions
Incident Categories
System Monitoring
Analytics & Reports
Notification Settings
System Settings
System Admin
admin@girs.com
Logout
Analytics & Reports
Platform-wide statistics and insights
Last 6 Months
Export Report
Total Incidents
3,847
+12.5% from last month
Resolution Rate
96.7%
+2.3% improvement
Active Users
1,284
+8.1% growth
Avg Response Time
3.9m
12% faster
Incident Trends by Category
Sep
fire : 22
medical : 48
flood : 18
crime : 16
accident : 35
Jul
Aug
Sep
Oct
Nov
Dec
fire
medical
flood
crime
accident
Incident Distribution by Category
Medical Emergency 29%
Accident 22%
Fire 15%
Flood 12%
Crime 8%
Others 13%
Average Response Time Trend
Barangay Performance
9:23 PM
4/7/2026

Chapter III

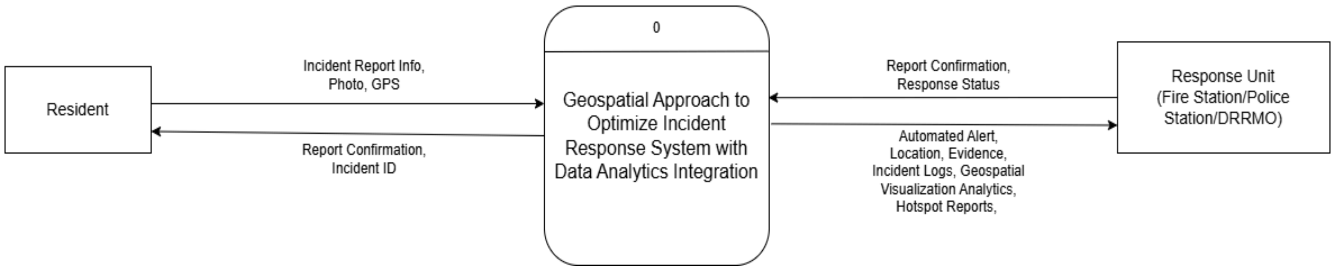
Methodology

This chapter introduces the systematic approach and tools used to develop and evaluate the Geospatial Approach to Optimize Incident Response System with Data Analytics Integration. The study adopts the Systems Development Life Cycle (SDLC) framework, which provides a structured process for planning, analyzing, designing, building, testing, and implementing the system. The chapter outlines the operational framework that guided the project and presents key modeling tools including the Context Diagram and Data Flow Diagrams (DFD) to illustrate system processes, the Entity-Relationship Diagram (ERD) to define data structures, and the Unified Modeling Language (UML) Use Case Diagram and scenarios to describe user interactions. A complete Data Dictionary was provided to define all data elements used in the system. The chapter also specified the necessary hardware and software requirements, presented the project timeline, and discussed the evaluation process, which includes system testing and assessment using the ISO/IEC 25010:2011 Software Quality Model to ensure functionality, reliability, and overall user experience.

Data Flow Diagram

Figure

Context Diagram of Geospatial Approach to Optimize Incident Response System with Data Analytics Integration



Figure

Data Flow Diagram (Level 1)

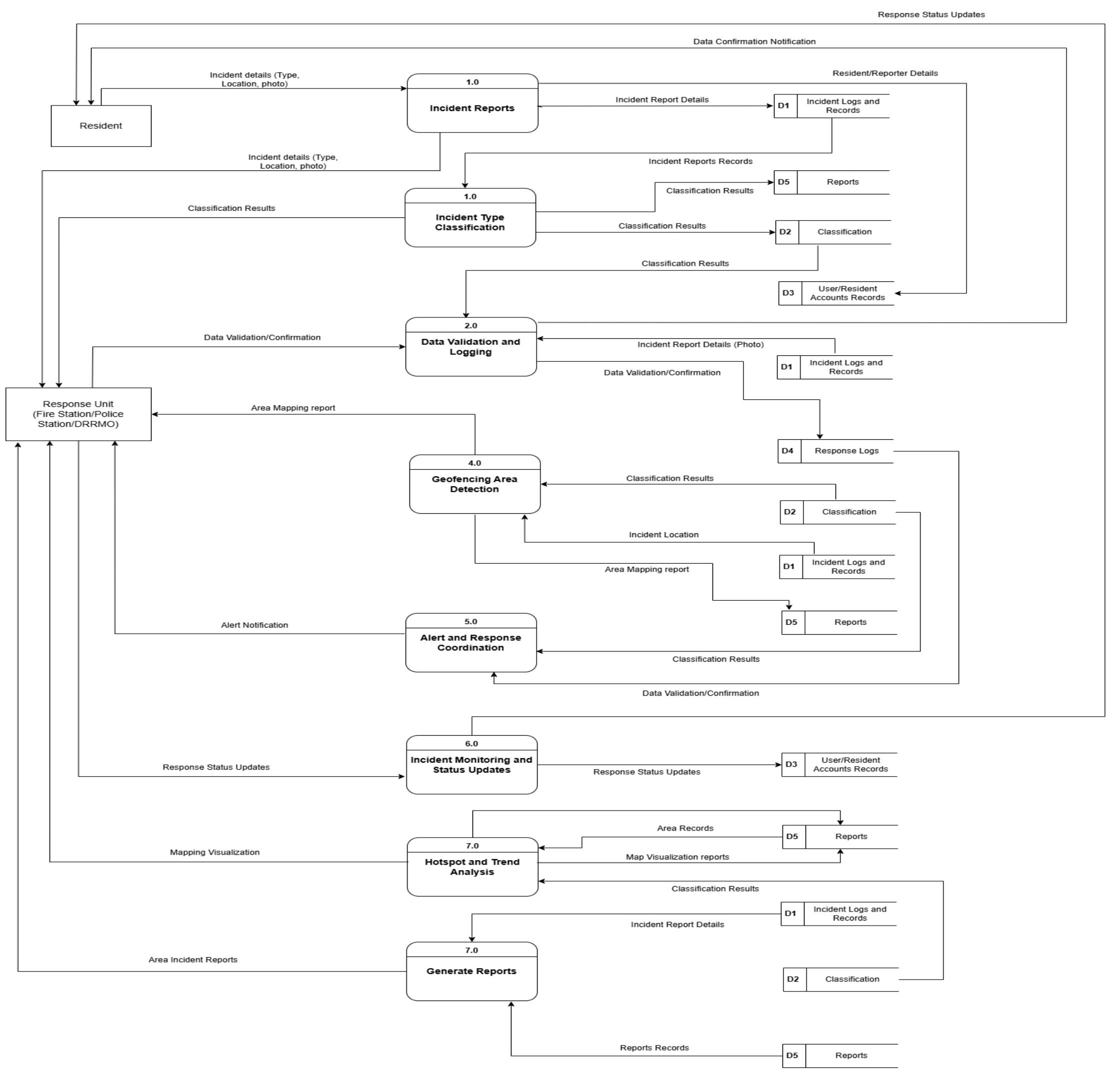


Figure #

Data Flow Diagram (Level 1)

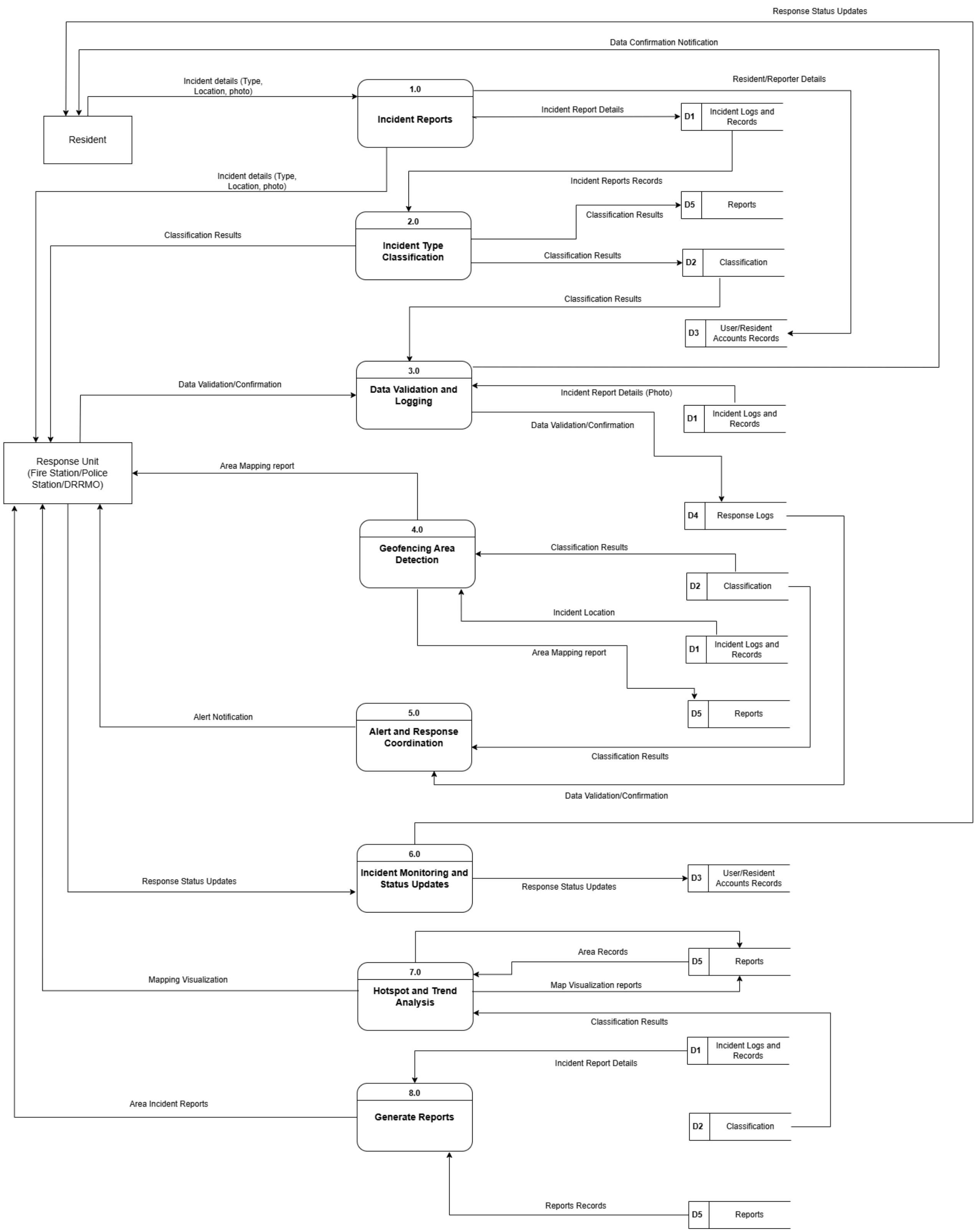
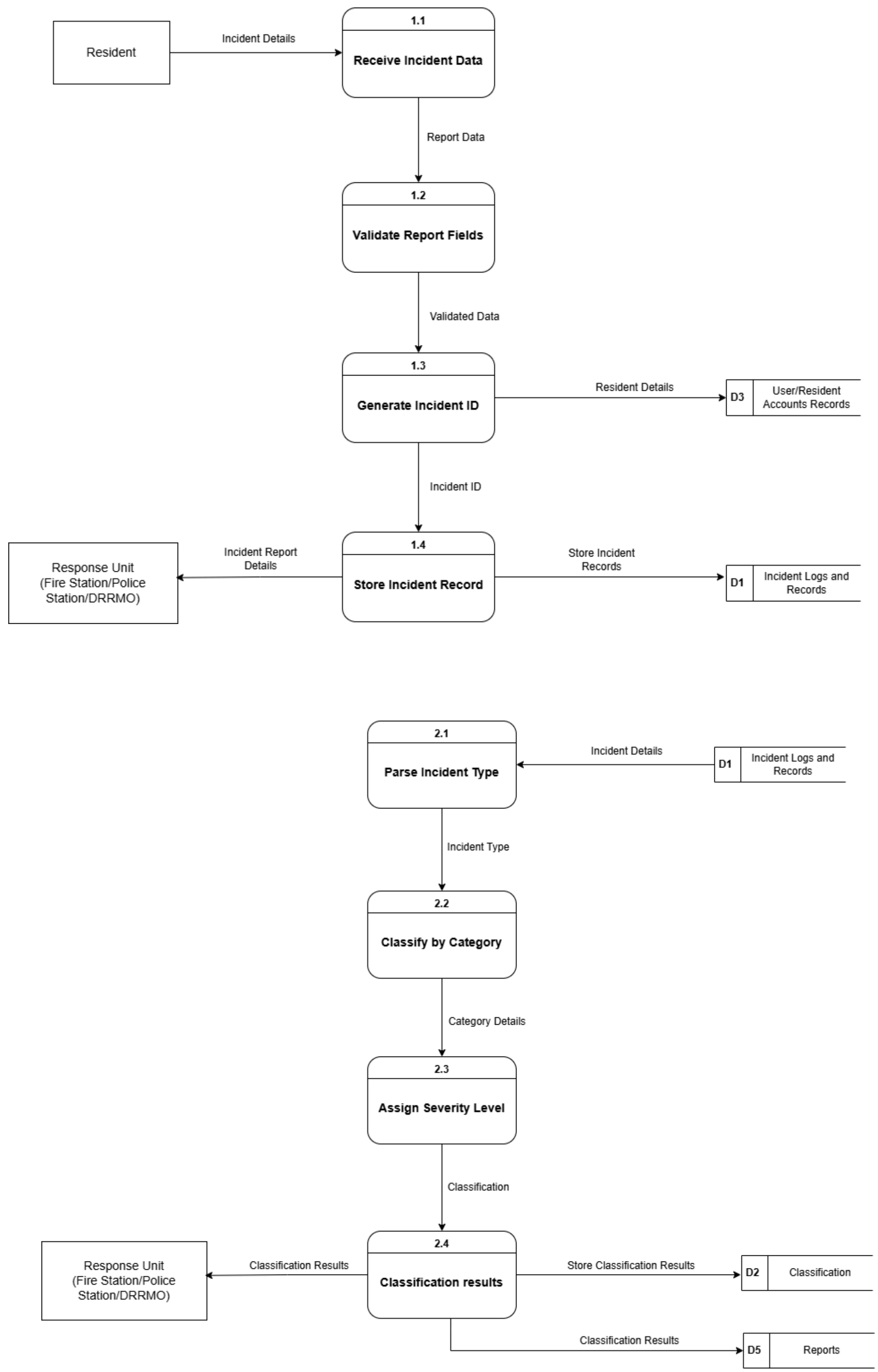
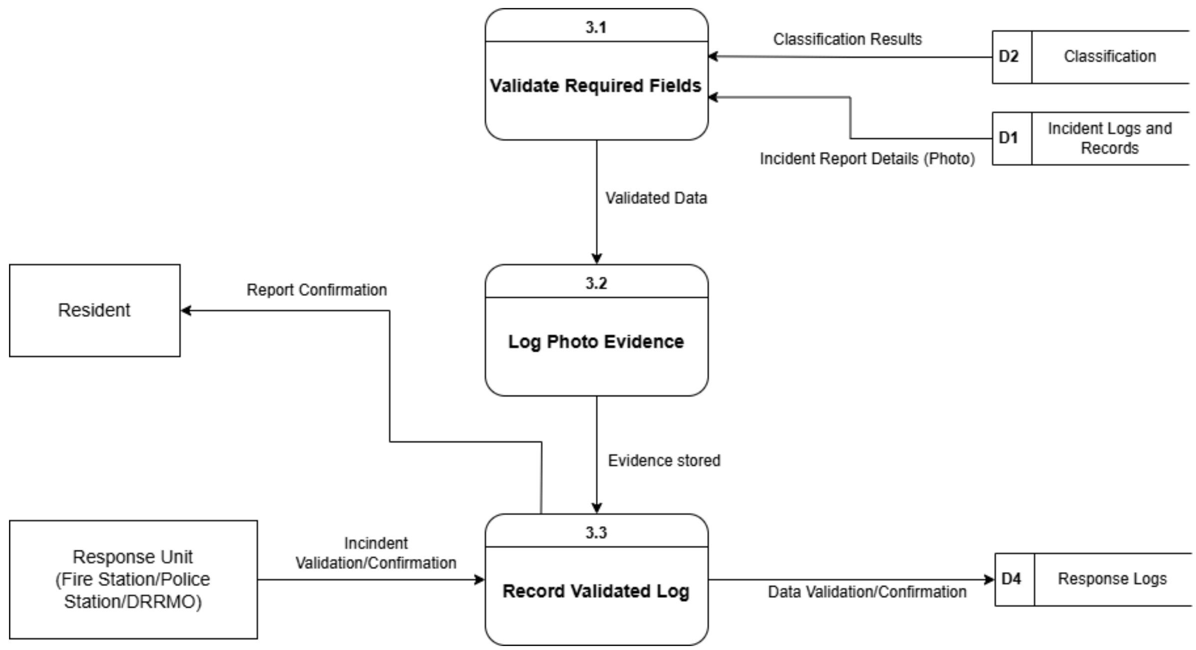
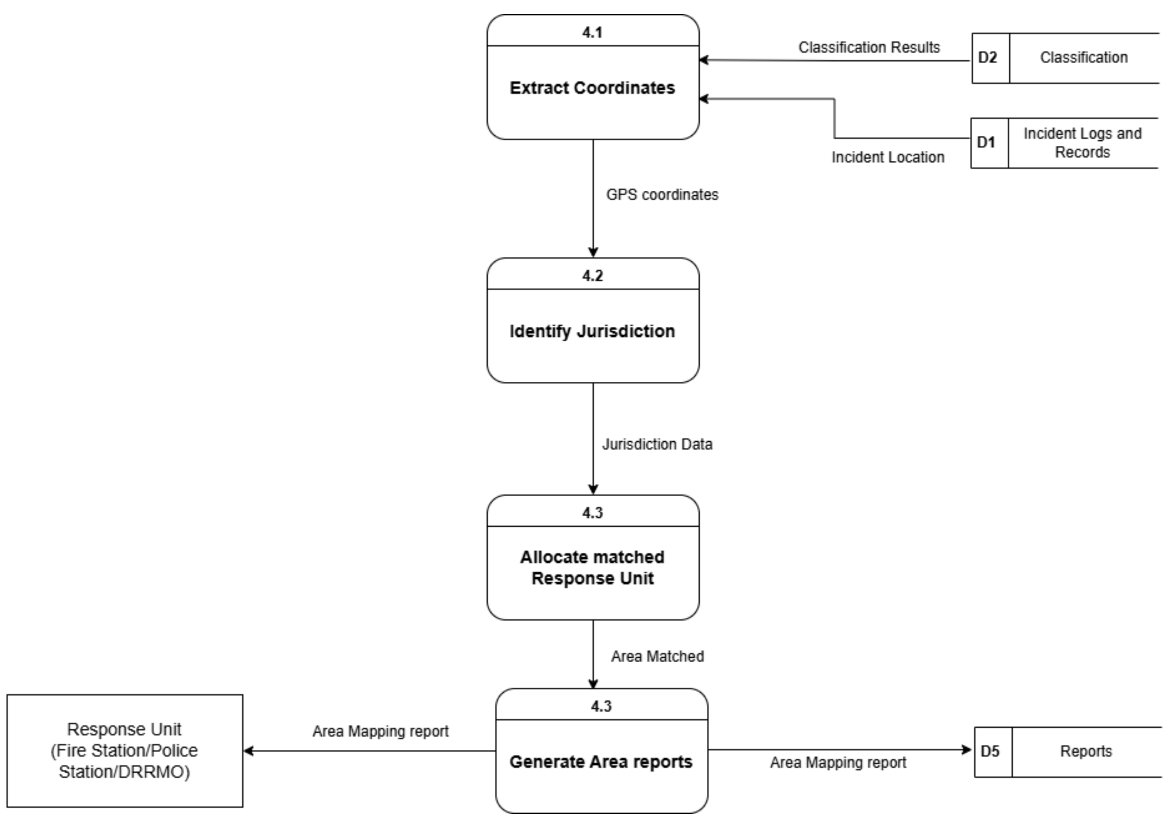


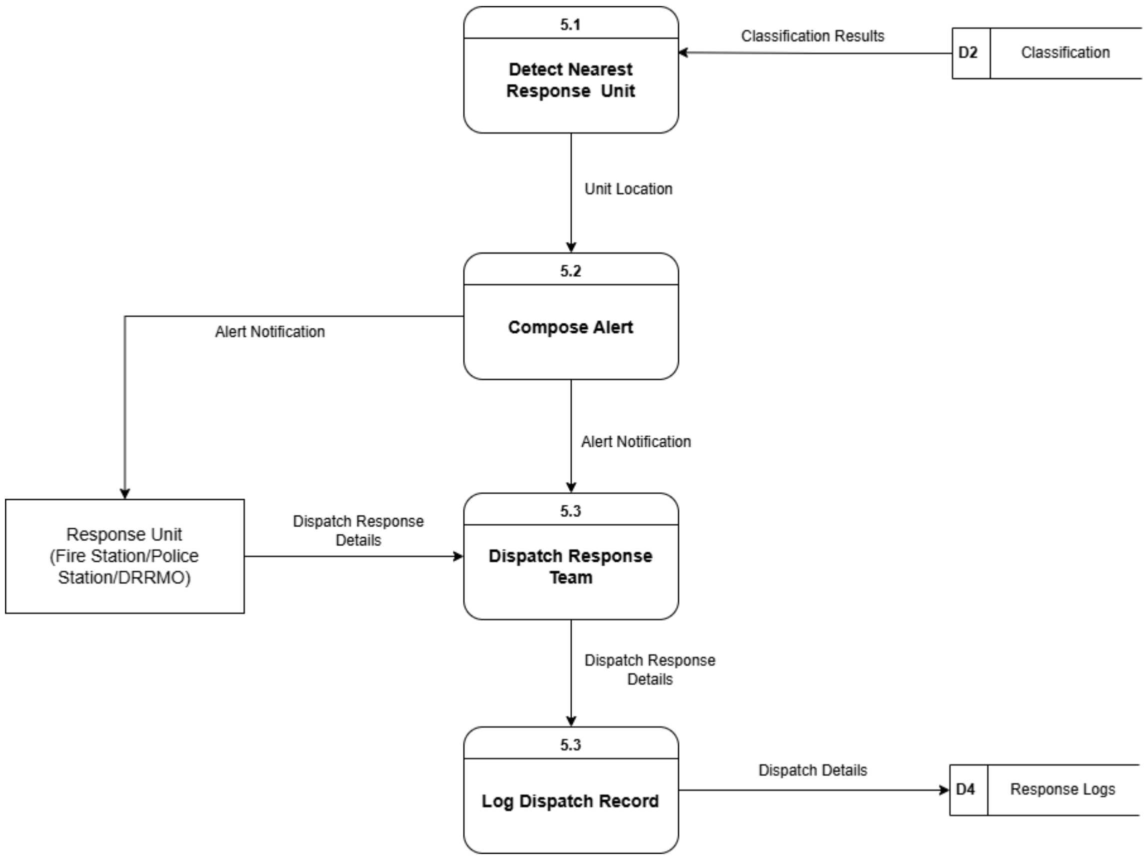
Figure #

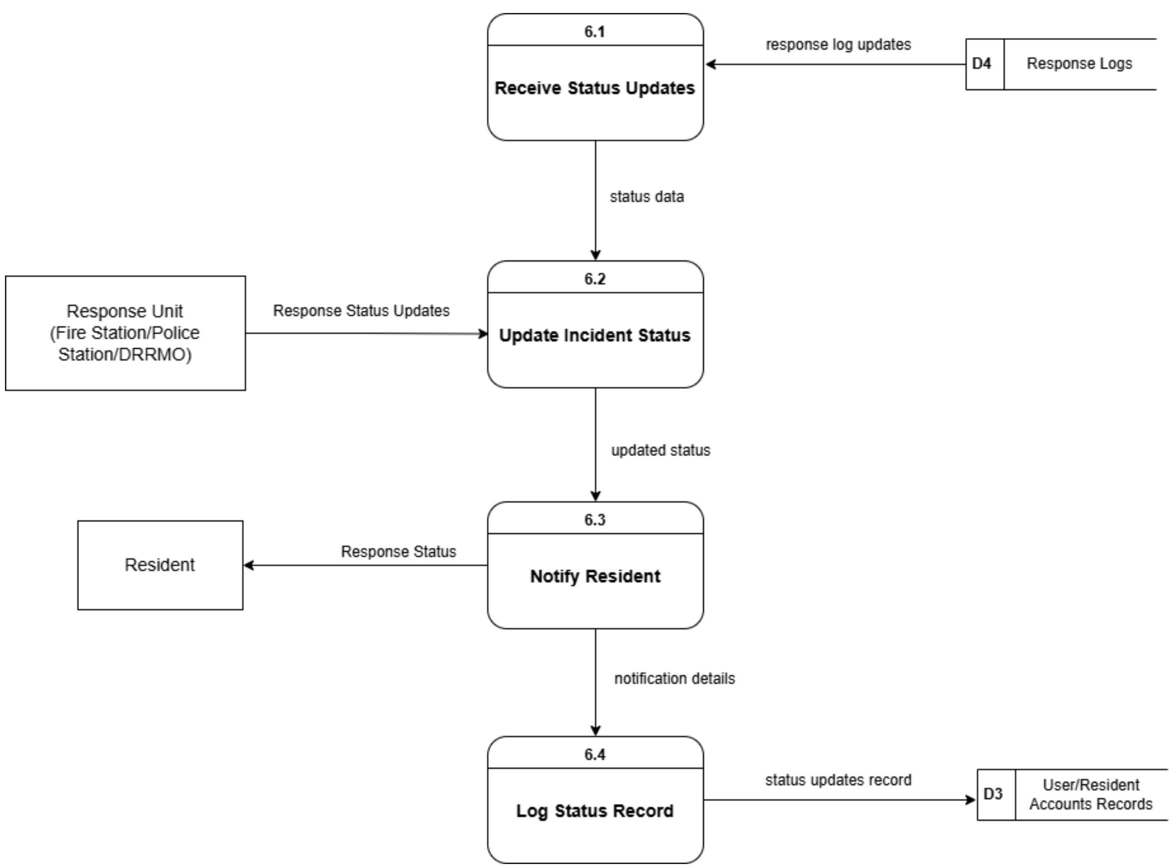
Data Flow Diagram (Level 2)

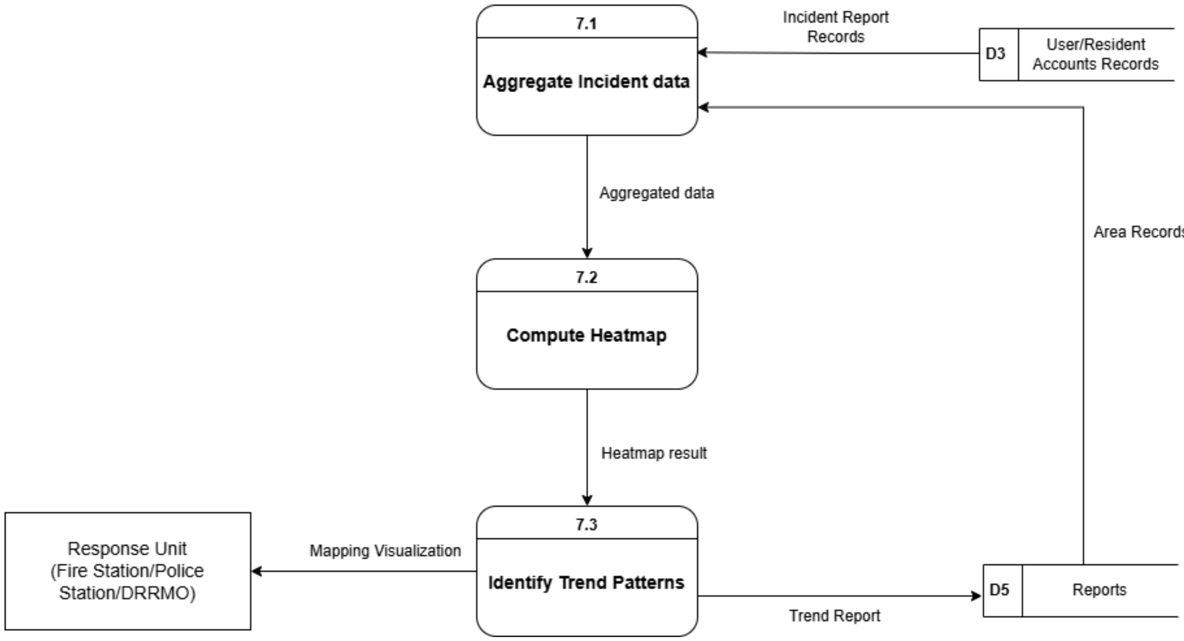


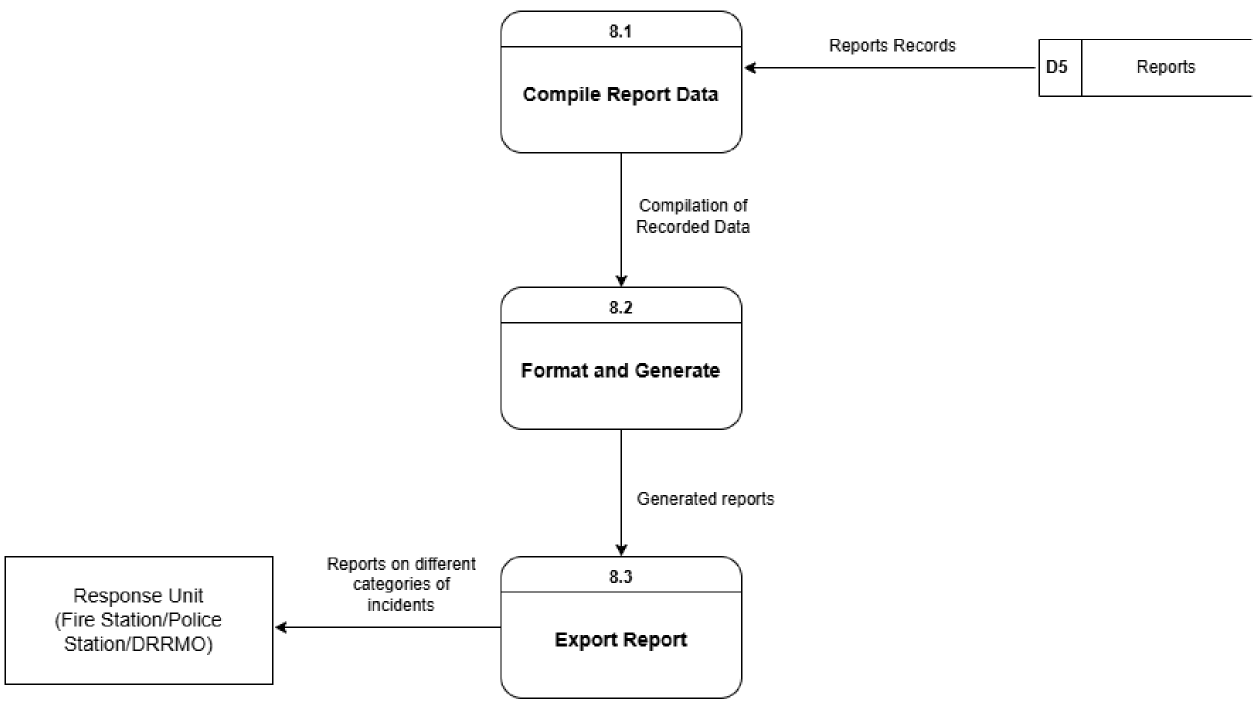

3.1
Validate Required Fields
Classification Results
D2
Classification
D1
Incident Logs and Records
Incident Report Details (Photo)
Validated Data
3.2
Log Photo Evidence
Resident
Report Confirmation
Evidence stored
3.3
Record Validated Log
Response Unit
(Fire Station/Police Station/DRRMO)
Incindent Validation/Confirmation
Data Validation/Confirmation
D4
Response Logs


4.1
Extract Coordinates
Classification Results
D2
Classification
D1
Incident Logs and Records
Incident Location
GPS coordinates
4.2
Identify Jurisdiction
Jurisdiction Data
4.3
Allocate matched Response Unit
Area Matched
4.3
Generate Area reports
Response Unit
(Fire Station/Police Station/DRRMO)
Area Mapping report
Area Mapping report
D5
Reports

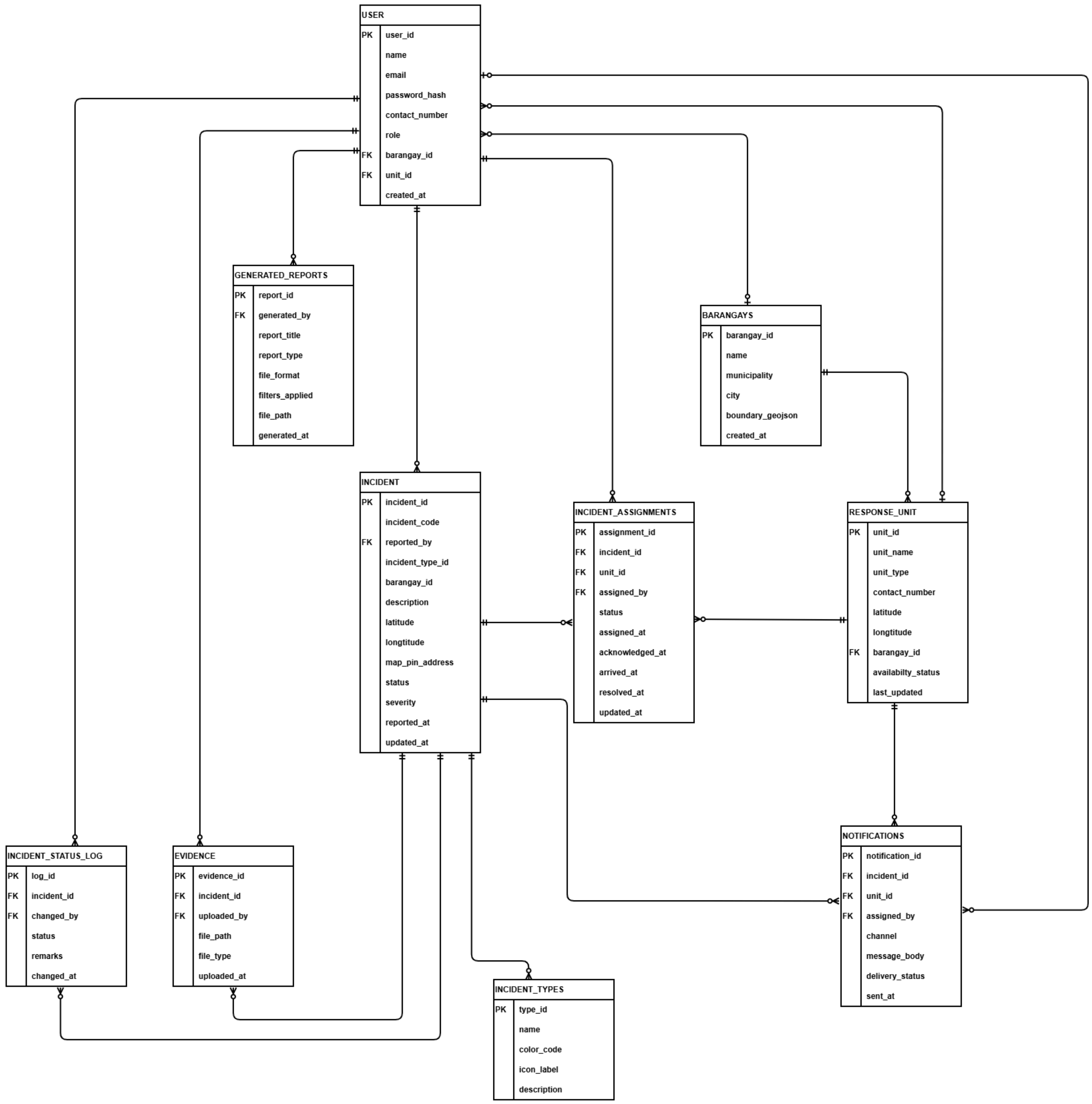

5.1
Detect Nearest Response Unit
Classification Results
D2
Classification
Unit Location
5.2
Compose Alert
Alert Notification
Alert Notification
Response Unit (Fire Station/Police Station/DRRMO)
Dispatch Response Details
5.3
Dispatch Response Team
Dispatch Response Details
5.3
Log Dispatch Record
Dispatch Details
D4
Response Logs


6.1
Receive Status Updates
response log updates
D4
Response Logs
status data
6.2
Update Incident Status
Response Unit (Fire Station/Police Station/DRRMO)
Response Status Updates
updated status
6.3
Notify Resident
Resident
Response Status
notification details
6.4
Log Status Record
status updates record
D3
User/Resident Accounts Records


7.1
Aggregate Incident data
Incident Report Records
D3
User/Resident Accounts Records
Aggregated data
Area Records
7.2
Compute Heatmap
Heatmap result
Response Unit (Fire Station/Police Station/DRRMO)
Mapping Visualization
7.3
Identify Trend Patterns
Trend Report
D5
Reports


8.1
Compile Report Data
Reports Records
D5
Reports
Compilation of Recorded Data
8.2
Format and Generate
Generated reports
Reports on different categories of incidents
Response Unit (Fire Station/Police Station/DRRMO)
8.3
Export Report

ERD



USER
PK user_id
name
email
password_hash
contact_number
role
FK barangay_id
FK unit_id
created_at
GENERATED_REPORTS
PK report_id
FK generated_by
report_title
report_type
file_format
filters_applied
file_path
generated_at
BARANGAYS
PK barangay_id
name
municipality
city
boundary_geojson
created_at
INCIDENT
PK incident_id
incident_code
FK reported_by
incident_type_id
barangay_id
description
latitude
longtitude
map_pin_address
status
severity
reported_at
updated_at
INCIDENT_ASSIGNMENTS
PK assignment_id
FK incident_id
FK unit_id
FK assigned_by
status
assigned_at
acknowledged_at
arrived_at
resolved_at
updated_at
RESPONSE_UNIT
PK unit_id
unit_name
unit_type
contact_number
latitude
longtitude
FK barangay_id
availabilty_status
last_updated
INCIDENT_STATUS_LOG
PK log_id
FK incident_id
FK changed_by
status
remarks
changed_at
EVIDENCE
PK evidence_id
FK incident_id
FK uploaded_by
file_path
file_type
uploaded_at
NOTIFICATIONS
PK notification_id
FK incident_id
FK unit_id
FK assigned_by
channel
message_body
delivery_status
sent_at
INCIDENT_TYPES
PK type_id
name
color_code
icon_label
description